

# Jae-Won Chung

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## Summary

I am a fifth year PhD candidate in CSE at the University of Michigan, advised by [Professor Mosharaf Chowdhury](#). I build **efficient software systems for machine learning**, with a focus on the efficient management of not only time, but also **energy**. My research views energy as a first-class systems resource that is worth carefully optimizing and allocating based on precise measurement and understanding.

I am passionate about **open-source** software and making real-world impact with my research. My open-source works, including the [Zeus](#) library, have received wide recognition from academia and industry from for instance NVIDIA, [Google](#), Microsoft, [PyTorch Foundation](#), and [GitHub](#). I created and lead the [ML.ENERGY initiative](#) as part of my research and open-source efforts, which is now a cross-institutional effort.

## Education

### University of Michigan

PH.D. CANDIDATE IN COMPUTER SCIENCE AND ENGINEERING

*Ann Arbor, MI, USA*

*Sep 2021 - present*

### University of Michigan

M.S. IN COMPUTER SCIENCE AND ENGINEERING

*Ann Arbor, MI, USA*

*Sep 2021 - Apr 2023*

### Seoul National University

B.S. IN ELECTRICAL AND COMPUTER ENGINEERING

*Seoul, South Korea*

*Mar 2015 - Aug 2021*

- GPA: 4.04/4.3 (overall) 4.15/4.3 (major), Summa Cum Laude. Period includes two years of military service.

## Publications

### Peer-reviewed conference publications

\* Equal contribution

- **Kareus: Joint Reduction of Dynamic and Static Energy in Large Model Training**, Ruofan Wu, [Jae-Won Chung](#), Mosharaf Chowdhury, **OSDI**, 2026 (Acceptance rate = 20.0%)
- **GPU-to-Grid: Voltage Regulation via GPU Utilization Control**, Zhirui Liang, [Jae-Won Chung](#), Mosharaf Chowdhury, Jiasi Chen, Vladimir Dvorkin, **PowerUp**, 2026
- **The ML.ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization**, [Jae-Won Chung](#), Jeff J. Ma, Ruofan Wu, Jiachen Liu, Oh Jun Kweon, Yuxuan Xia, Zhiyu Wu, Mosharaf Chowdhury, **NeurIPS Datasets & Benchmarks track (spotlight)**, 2025 (Spotlight acceptance rate = 2.81%)
- **Reducing Energy Bloat in Large Model Training**, [Jae-Won Chung](#), Yile Gu, Insu Jang, Luoxi Meng, Nikhil Bansal, Mosharaf Chowdhury, **SOSP**, 2024 (Acceptance rate = 17.34%)
- **Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training**, Jie You\*, [Jae-Won Chung](#)\*, Mosharaf Chowdhury, **NSDI**, 2023 (Acceptance rate = 18.38%)
- **ShadowTutor: Distributed Partial Distillation for Mobile Video DNN Inference**, [Jae-Won Chung](#), Jae-Yun Kim, Soo-Mook Moon, International Conference on Parallel Processing (**ICPP**), 2020 (Acceptance rate = 28.99%)

### Preprints, technical reports, and workshop publications

- **OpenG2G: A Simulation Platform for AI Datacenter-Grid Runtime Coordination**, [Jae-Won Chung](#)\*, Zhirui Liang\*, Yanyong Mao, Jiasi Chen, Mosharaf Chowdhury, Vladimir Dvorkin, 2026
- **Cornserve: A Distributed Serving System for Any-to-Any Multimodal Models**, [Jae-Won Chung](#)\*, Jeff J. Ma\*, Jisang Ahn, Yizhuo Liang, Akshay Jajoo, Myungjin Lee, Mosharaf Chowdhury, **ACM CAIS Demos**, 2026
- **Where Do the Joules Go? Diagnosing Inference Energy Consumption**, [Jae-Won Chung](#), Ruofan Wu, Jeff J. Ma, Mosharaf Chowdhury, 2026
- **Cornserve: Efficiently Serving Any-to-Any Multimodal Models**, Jeff J. Ma\*, [Jae-Won Chung](#)\*, Akshay Jajoo, Myungjin Lee, Mosharaf Chowdhury, 2025

- **Toward Cross-Layer Energy Optimizations in AI Systems**, [Jae-Won Chung](#), Nishil Talati, Mosharaf Chowdhury, **DOE ASCR Energy-Efficient Computing for Science Workshop**, 2024
- **Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services**, Jiachen Liu, [Jae-Won Chung](#), Zhiyu Wu, Fan Lai, Myungjin Lee, Mosharaf Chowdhury, 2024
- **Chasing Low-Carbon Electricity for Practical and Sustainable DNN Training**, Zhenning Yang, Luoxi Meng, [Jae-Won Chung](#), Mosharaf Chowdhury, **ICLR Workshop: Tackling Climate Change with Machine Learning**, 2023

## Open Source Projects

- **Zeus** (☆356 ♪ 45), A framework for deep learning energy measurement and optimization. **A PyTorch ecosystem project. GitHub secure open source fund cohort.** Provides a unified abstraction over various CPU, GPU, and SoC vendors, and an array of time & energy optimizers.
- **Cornserve** (☆124 ♪ 10), An efficient serving platform for any-to-any multimodal models. Takes a microservice approach to serving complex-structured multimodal input & output models.
- **OpenG2G** (☆9 ♪ 0), Modular simulation library for AI datacenter-grid interaction. Enables research on how AI datacenters interact with the power grid through composable, extensible simulation modules.
- **BERT4Rec-VAE-Pytorch** (☆415 ♪ 99), A PyTorch framework for recommendation model training, with abstract classes for pluggable model, dataset, and samplers. BERT4Rec and Netflix VAE models implemented.
- **Reason** (☆197 ♪ 4), A shell for managing research papers, written in Rust. Supports importing papers from file and URL, attaching markdown notes, and creating an HTML book with notes. Uses *serde* to persist data in human-readable and cloud sync-friendly format.
- **Pegasus** (☆33 ♪ 3), An SSH command runner with a focus on simplicity, written in Rust. Runs multiple commands asynchronously using the *tokio* runtime and streams stdout and stderr back to the user. Battle-tested through multiple research projects and benchmarking.

Number of stars and forks are up-to-date as of May 7th, 2026.

## Honors & Awards

|          |                                                                                                          |                                         |
|----------|----------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Apr 2026 | <b>MLSys Rising Stars</b> , MLCommons Machine Learning and Systems Rising Stars 2026 cohort              | <i>MLCommons</i>                        |
| Nov 2025 | <b>Slingshot Grant</b> , \$25,000 USD AWS Credits for the development of the <u>ML.ENERGY Initiative</u> | <i>Laude Institute</i>                  |
| Aug 2025 | <b>GitHub Secure Open Source Fund</b> , \$10,000 for the development of the <u>Zeus</u> project          | <i>GitHub</i>                           |
| May 2024 | <b>PyTorch Ecosystem Project</b> , <u>Zeus</u> was included in the PyTorch Ecosystem                     | <i>PyTorch Foundation</i>               |
| Jan 2024 | <b>Research award</b> , \$20,000 for the development of the <u>ML.ENERGY Initiative</u>                  | <i>Salesforce</i>                       |
| Jan 2024 | <b>Mozilla Technology Fund 2024</b> , \$50,000 for the development of the <u>Zeus</u> project            | <i>Mozilla</i>                          |
| Nov 2022 | <b>Carbon Hack '22 Second Best Solution</b> , <u>Carbon-Aware DNN Training with Zeus</u> , \$25,000      | <i>Green Software Foundation</i>        |
| Jul 2021 | <b>Kwanjeong Overseas Scholarship</b> , \$25,000                                                         | <i>Kwanjeong Educational Foundation</i> |
| Mar 2019 | <b>Kwanjeong Undergraduate Scholarship</b> , \$20,000 over two years                                     | <i>Kwanjeong Educational Foundation</i> |

## Talks

|          |                                                                                    |                                              |
|----------|------------------------------------------------------------------------------------|----------------------------------------------|
| Apr 2026 | <b>Energy-Efficient Machine Learning Systems</b>                                   | <i>UC Berkeley eMERGE Seminar</i>            |
| Dec 2025 | <b>Energy and Power as First-Class ML Design Metrics (w/ NVIDIA, Google, Meta)</b> | <i>NeurIPS 25 Tutorial</i>                   |
| Oct 2025 | <b>Energy as a First-Class ML Design Metric</b>                                    | <i>UW-Madison MadSystems Seminar</i>         |
| Sep 2025 | <b>Power and Energy as First-Class AI Design Metrics</b>                           | <i>KPAI (Bay Area Korean AI Meetup)</i>      |
| Jun 2025 | <b>Energy as a First-Class Resource in Machine Learning Systems</b>                | <i>Pruna AI</i>                              |
| May 2025 | <b>Energy-Efficient Systems for Machine Learning</b>                               | <i>Harvard Power and AI Initiative</i>       |
| Nov 2024 | <b>Energy-Efficient Systems for Machine Learning</b>                               | <i>SOSP 24 Doctoral Workshop</i>             |
| Apr 2024 | <b>Power and Energy Considerations in Machine Learning Systems</b>                 | <i>University of Michigan (EECS 598)</i>     |
| Oct 2023 | <b>Energy-Efficient Software Systems for Machine Learning</b>                      | <i>Seoul National University</i>             |
| Oct 2023 | <b>Energy-Efficient Deep Learning with PyTorch and Zeus</b>                        | <i>PyTorch Conference</i>                    |
| Sep 2023 | <b>Energy-Efficient Deep Learning with Zeus</b>                                    | <i>Massachusetts Institute of Technology</i> |

## Selected Media Coverage

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My research and open-source works have been covered by various media outlets, including MIT Technology Review, ArsTechnica, and Science News.

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|----------|----------------------------------------------------------------------------------------------------------------|---------------------------------|
| Jul 2025 | <u><a href="#">How much energy does your AI prompt use? It depends.</a></u>                                    | <i>Science News</i>             |
| May 2025 | <u><a href="#">We did the math on AI's energy footprint. Here's the story you haven't heard.</a></u>           | <i>MIT Technology Review</i>    |
| May 2025 | <u><a href="#">AI Consumes Lots of Energy. Can It Ever Be Sustainable?</a></u>                                 | <i>The New Stack</i>            |
| Mar 2025 | <u><a href="#">Can we make AI less power-hungry? These researchers are working on it.</a></u>                  | <i>ArsTechnica (front page)</i> |
| Nov 2024 | <u><a href="#">Up to 30% of the power used to train AI is wasted: A software tool could help fix that.</a></u> | <i>Tech Xplore</i>              |
| Apr 2023 | <u><a href="#">University of Michigan's 'Zeus' Framework Downsizes AI's Massive Carbon Footprint.</a></u>      | <i>HPCWire</i>                  |
| Apr 2023 | <u><a href="#">Researchers claim they can cut AI training energy demands by 75%.</a></u>                       | <i>DatacenterDynamics</i>       |

## Service

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- **Ad-hoc reviewer**, Nature
- **Systems Reading Group Organizer**, Paper reading group inside Michigan CSE. Organizer from September 2022 to December 2025.

## Teaching

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- **Systems for Generative AI (UMich, Fall 25)**, GSI, three lectures on GenAI and GenAI systems fundamentals. **Instructor evaluation median 5.00 (highest possible)**. Written comments included phrases “excellent” (three times), “perfect,” and “one of the best.”
- **Operating Systems (SNU, Spring 21)**, Lead TA, managed Linux kernel hacking projects and led student team design reviews.
- **Computer Architecture (SNU, Fall 20)**, Peer tutor, provided 30 hours of online lecture. **Best Tutor Award**.